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Q-MECH: A GPU-Accelerated Open-Source Python Pipeline for Quantum Mechanical (DFT & QM/MM) Analysis of Molecular Mechanism of Inhibition in Protein–Ligand Complexes

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Software

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Statement of Need

Understanding the molecular mechanism by which small molecules inhibit protein targets is central to rational drug design. While molecular docking and molecular dynamics (MD) simulations provide structural insights into binding modes, they lack the quantum mechanical description necessary to characterize covalent interactions, electronic reactivity, and electrostatic complementarity at the electronic structure level (Ryde & Söderhjelm, 2016; Sherrill, 2013).

Density functional theory (DFT) methods offer rigorous access to frontier molecular orbital energies, global chemical reactivity descriptors (ionization potential, electron affinity, electronegativity, chemical hardness, softness, and electrophilicity index), Fukui reactivity indices, and molecular electrostatic potential (MEP) that govern protein–ligand recognition (Parr & Yang, 1995; Yang & Mortier, 1986). Hybrid quantum mechanics/molecular mechanics (QM/MM) approaches further enable calculation of interaction energies with explicit inclusion of the protein electrostatic environment (Warshel & Levitt, 1976).

However, the practical application of these methods to drug discovery workflows has been hindered by: (i) steep learning curves and lack of automation, (ii) high computational cost, (iii) dependence on expensive commercial software (e.g., Gaussian, ORCA), and (iv) absence of integrated, user-friendly pipelines that bridge quantum chemistry with structural bioinformatics. Existing tools either require SMILES input (losing binding conformation information), lack QM/MM capability, or do not support GPU acceleration on consumer hardware.

Q-MECH addresses these gaps by providing a fully automated, open-source Python pipeline that takes a single protein–ligand complex PDB file as the only required input and produces a comprehensive quantum chemical characterization of the inhibition mechanism, including publication-ready figures and structured data reports. GPU acceleration via GPU4PySCF (Wu et al., 2024) reduces computation time from hours to approximately 50 minutes per complex on a single consumer-grade GPU, making production-grade DFT calculations accessible to the broader drug discovery community.

Summary

Q-MECH is a Python pipeline that integrates twelve analysis steps into a single automated workflow:

1. **Binding pocket extraction** from PDB using Biopython (Cock et al., 2009), with automatic ligand detection and AMBER ff14SB (Maier et al., 2015) charge assignment to pocket

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residues (default cutoff: 8.0 Å).

2. **Ligand preparation** directly from PDB coordinates — no SMILES required — with molecular connectivity inferred from van der Waals radii and bond orders determined by RDKit (Bento et al., 2020; Landrum, 2013).
3. **Real DFT calculations** at the B3LYP/def2-SVP level (Becke, 1993; Lee et al., 1988; Weigend & Ahlrichs, 2005) using PySCF (Sun et al., 2018, 2020) with optional GPU acceleration via GPU4PySCF (Wu et al., 2024). Basis set is automatically selected based on ligand heavy atom count (def2-SVP for 60 atoms, STO-3G fallback for convergence failures).
4. **Real frontier molecular orbital energies** (HOMO, LUMO) from the converged Kohn–Sham wavefunction, with global chemical reactivity descriptors via Koopmans' theorem (Koopmans, 1934): ionization potential ($IP = -E_{HOMO}$), electron affinity ($EA = -E_{LUMO}$), electronegativity (χ), chemical hardness (η), global softness (S), and electrophilicity index (ω).
5. **Real Fukui reactivity indices** from three separate DFT calculations on the neutral (N), anionic (N+1), and cationic (N−1) electronic states (Yang & Mortier, 1986), condensed to atoms via Mulliken population analysis: f (electrophilic attack sites), f (nucleophilic attack sites), f (radical attack sites), and the dual descriptor $\Delta f = f - f$.
6. **Real molecular electrostatic potential** computed from the DFT electron density on a three-dimensional grid surrounding the van der Waals surface, including both nuclear and electronic contributions.
7. **QM/MM electrostatic embedding** via PySCF's qmmm module (pyscf.qmmm.mm_charge (Sun et al., 2018)), modifying the QM Hamiltonian with MM point charges: $\hat{H}_{QM/MM} = \hat{H}_{QM} + \Sigma_M\ Q_M/|\hat{r} - R_M|$. Total interaction energy includes the embedded QM energy, classical Coulomb interaction, and Lennard-Jones van der Waals term.
8. **Per-residue energy decomposition** partitioning electrostatic and van der Waals contributions per binding pocket residue, enabling identification of pharmacophoric binding interactions.
9. **Protein–ligand hydrogen bond analysis** using geometric criterion (donor–acceptor distance 3.5 Å) with Mulliken charge-based donor/acceptor assignment, classified as strong (<3.0 Å), moderate (3.0–3.2 Å), or weak (3.2–3.5 Å) (Jeffrey, 1997).
10. **Rule-based mechanism interpretation** classifying inhibition mode (covalent/irreversible, competitive, allosteric), reactivity class, electrophilicity, metal chelation potential, and selectivity from computed quantum descriptors.
11. **Drug-likeness assessment** using Lipinski's Rule of Five (Lipinski et al., 2001), Quantitative Estimate of Drug-likeness (QED) (Bickerton et al., 2012), and topological polar surface area (TPSA), computed via RDKit (RDKit, n.d.).
12. **Automated publication-quality figure generation** — eight 600 DPI PNG figures per compound plus a multi-compound comparative figure — using Matplotlib (Hunter, 2007).

Implementation

Q-MECH is implemented in Python 3.10+ and organized as a single-file pipeline (`qmech_v1.0.py`) with a companion batch runner (`batch_run_qmech.sh`) and figure generator (`qmech_figures.py`). All dependencies are freely available via pip.

The pipeline supports both GPU (via GPU4PySCF, CUDA 11.x/12.x) and CPU execution, with automatic fallback to CPU mode when GPU is unavailable. Multiple complexes can be

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processed sequentially in batch mode, with each complex producing an independent result directory containing JSON and plain-text reports alongside all figures.

Performance

On an NVIDIA GeForce RTX 3070 Ti (8 GB VRAM, CUDA 11.8) with an Intel 24-thread CPU and 128 GB RAM, Q-MECH processes a typical drug-like compound (30–50 heavy atoms) in approximately 65 minutes: DFT SCF (~5 min GPU), Fukui N/N+1/N−1 (~15 min GPU), QM/MM (~45 min CPU), and figure generation (~2 min). CPU-only execution requires approximately 3–4× longer. Note that the QM/MM step runs on CPU as GPU4PySCF does not currently support the `pyscf.qmmm.mm_charge` interface.

Output

For each compound, Q-MECH produces the following files in the output directory:

File	Content
`fig1_homo_lumo.png`	HOMO/LUMO orbital energy diagram
`fig2_dft_radar.png`	DFT reactivity radar chart (5 descriptors)
`fig3_fukui.png`	Fukui f /f /Δf indices with reactive site annotations
`fig4_mep.png`	MEP surface feature bar chart
`fig5_per_residue.png`	Per-residue QM/MM energy decomposition
`fig6_elec_vs_lj.png`	Electrostatic vs van der Waals contributions
`fig7_hbonds.png`	Protein–ligand hydrogen bond distance chart
`fig8_mechanism.png`	Inhibition mechanism summary table
`report.json`	Machine-readable structured results
`report.txt`	Human-readable text report

AI Disclosure

Portions of the Q-MECH source code and manuscript text were drafted with the assistance of Claude (Anthropic, claude.ai), an AI language model. All AI-generated content was critically reviewed, validated, and edited by the author. The scientific design, conceptualization, validation, and interpretation of results are entirely the work of the author. The computational results reported herein were generated by running the pipeline on real protein–ligand complex data, and all quantum chemical calculations were performed using established open-source software (PySCF, GPU4PySCF) without AI involvement in the actual calculations.

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